Degenerate Perturbation Theory

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October 17, 2024

If the unperturbed states are degenerate – that is, if the distinct states ψ_a^0 and ψ_b^0 share the same energy, then ordinary perturbation theory fails. We derived the equations for the first-order correction to the wave function,

$$\psi_n^1 = \sum_{m \neq n} \frac{\langle \psi_m^0 | H' | \psi_n^0 \rangle}{E_n^0 - E_m^0} \psi_m^0,$$

and the second-order correction to the energy,

$$E_n^2 = \sum_{m \neq n} \frac{|\langle \psi_m^0 | H' | \psi_n^0 \rangle|^2}{E_n^0 - E_m^0}.$$

Of course, if calculating ψ_a^1 or E_a^2, these equations blow up. In the degenerate case, thus, there is no reason to trust even the *first* order correction to the energy. Almost all applications of perturbation theory involves degeneracy.

1 Two-Fold Degeneracy

Suppose we have two orthonormal solutions to the Schrödinger equation that share the same energy eigenvalue,

$$H^0 \psi_a^0 = E^0 \psi_a^0, \quad H^0 \psi_b^0 = E^0 \psi_b^0, \quad \langle \psi_a^0 | \psi_b^0 \rangle = 0, \tag{1}$$

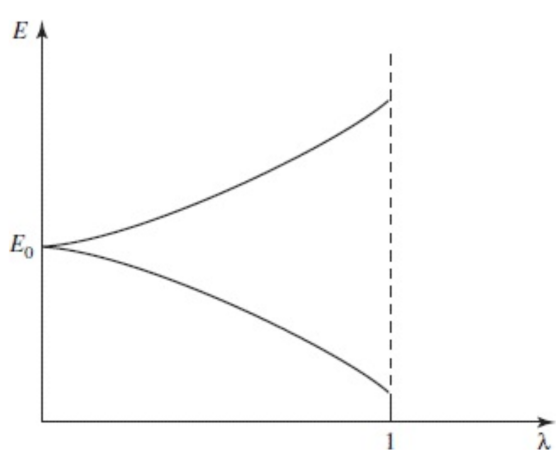
with ψ_a^0 and ψ_b^0 being normalized. Any linear combination of these states

$$\psi^0 = \alpha \psi_a^0 + \beta \psi_b^0$$

is also an eigenstate of H^0, with the same eigenvalue:

$$H^0 \psi^0 = E^0 \psi^0.$$

Perturbation (H'), however, breaks (or "lifts") the degeneracy. As we increase λ from 0 to 1, (unperturbed to perturbed), the common unperturbed energy E^0 splits into two.



Going in the other direction, when we "turn off" the perturbation, the *upper* state reduces down to one linear combination of ψ_a^0 and ψ_b^0, and the *lower* state reduces to some orthogonal linear combination, but we don't know beforehand what these *good* linear combinations are. For this reason we can't evaluate the first-order energy correction either – we don't know what unperturbed states to use.
Put simply, because the original states were degenerate, it's unclear which combination of those degenerate states we should use to represent the "new" states after the perturbation is applied.
The *good* states are defined as the limit of the true eigenstates as the perturbation is switched off ($\lambda \to 0$), but realistically, if you *knew* the exact eigenstates, you wouldn't need perturbation theory, so what do we do?
Let's first write the "good" unperturbed states in a generic form,

$$\psi^0 = \alpha\psi_a^0 + \beta\psi_b^0 \tag{2}$$

keeping α and β adjustable. We want to solve the Schrödinger equation

$$H\psi = E\psi \tag{3}$$

with $H = H^0 + \lambda H'$ and

$$E = E^0 + \lambda E^1 + \lambda^2 E^2 + \cdots, \quad \psi = \psi^0 + \lambda\psi^1 + \lambda^2\psi^2 + \cdots.$$

Plugging into Equation 3, and collecting like powers of λ like before, we get

$$H^0\psi^0 + \lambda\left(H'\psi^0 + H^0\psi^1\right) + \cdots = E^0\psi^0 + \lambda\left(E^1\psi^0 + E^0\psi^1\right) + \cdots,$$

Cancelling out $H^0\psi^0 = E^0\psi^0$, to order λ^1, we get

$$H^0\psi^1 + H'\psi^0 = E^0\psi^1 + E^1\psi^0. \tag{4}$$

Now we take the inner product with $\hat{\psi_a^0}$,

$$\left\langle\psi_a^0|H^0\psi^1\right\rangle + \left\langle\psi_a^0|H'\psi^0\right\rangle + E^0\left\langle\psi_a^1|\psi^1\right\rangle + E^1\left\langle\psi_a^0|\psi^0\right\rangle.$$

And again, since H^0 is hermitian, the first term on the left cancels the first term on the right. Plugging in our generic "good" unperturbed state (Equation 2) and exploiting orthonormality (Equation 1), we obtain

$$\alpha\left\langle\psi_a^0|H'|\psi_a^0\right\rangle + \beta\left\langle\psi_a^0|H'|\psi_b^0\right\rangle = \alpha E^1$$

or, more compactly,

$$\alpha W_{aa} + \beta W_{ab} = \alpha E^1, \tag{5}$$

where

$$W_{ij} \equiv \left\langle\psi_i^0|H'|\psi_j^0\right\rangle. \tag{6}$$

The Ws are known – they are just the matrix elements of H' with respect to the unperturbed wave functions ψ_a^0 and ψ_b^0. Similarly, taking the inner product with ψ_b^0 yields

$$\alpha W_{ba} + \beta W_{bb} = \beta E^1 \tag{7}$$

In matrix form, Equations 5 and 7 become

$$\begin{pmatrix} W_{aa} & W_{ab} \\ W_{ba} & W_{bb} \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = E^1 \begin{pmatrix} \alpha \\ \beta \end{pmatrix}. \tag{8}$$

Therefore, W's eigenvalues give the first order energy correction E^1, and the corresponding eigenvectors tell us the coefficients α and β that give us the "good" states. Solving for the eigenvalues yields

First-order energy eigenvalues

$$E^1_\pm = \frac{1}{2}\left[W_{aa} + W_{bb} \pm \sqrt{(W_{aa} - W_{bb})^2 + 4|W_{ab}|^2}\right]. \tag{9}$$

This is the fundamental result of degenerate perturbation theory. The two roots correspond to the two split perturbed energies.
If it happens that $W_{ab} = 0$, then the two eigenvectors are

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ and } \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

and their energies,

$$E^1_+ = W_{aa} = \langle \psi^0_a | H' | \psi^0_a \rangle, \quad E^1_- = W_{bb} = \langle \psi^0_b | H' | \psi^0_b \rangle$$

are precisely what we would have obtained via *non*degenerate perturbation theory.

2 "Good" States

I provide a theorem that allows one to *guess* "good" states. This allows us to go ahead and just use ordinary *non*degenerate perturbation theory.

Theorem: Let A be a hermitian operator that commutes with H^0 and H'. If ψ^0_a and ψ^0_b (the degenerate eigenfunctions of H^0 are also eigenfunctions of A, with distinct eigenvalues,

$$A\psi^0 = \mu\psi^0_a, \quad A\psi^0_b = \nu\psi^0_b, \quad \mu \neq \nu,$$

then ψ^0_a and ψ^0_b are "good" states to use in perturbation theory.

Proof: Since $H(\lambda) = H^0 + \lambda H'$ and A commute, there exist simultaneous eigenstates $\psi_\gamma(\lambda)$ where

$$H(\lambda)\psi_\gamma(\lambda) = E(\lambda)\psi_\gamma(\lambda) \text{ and } A\psi_\gamma(\lambda) = \gamma\psi_\gamma(\lambda).$$

The fact the A is hermitian means

$$\begin{aligned} \langle \psi^0_a | A\psi_\gamma(\lambda) \rangle &= \langle A\psi^0_a | \psi_\gamma(\lambda) \rangle \\ \gamma \langle \psi^0_a | \psi_\gamma(\lambda) \rangle &= \mu^* \langle \psi^0_a | \psi_\gamma(\lambda) \rangle \\ (\gamma - \mu) \langle \psi^0_a | \psi_\gamma(\lambda) \rangle &= 0 \end{aligned}$$

making use of the fact that μ is real. This holds true for any value of λ and taking the limit as $\lambda \to 0$, we have

$$\left\langle \psi_a^0 | \psi_\gamma(0) \right\rangle = 0 \quad \text{unless} \quad \gamma = \mu,$$

and similarly,

$$\left\langle \psi_b^0 | \psi_\gamma(0) \right\rangle = 0 \quad \text{unless} \quad \gamma = \nu.$$

Now the good states are linear combinations of ψ_a^0 and ψ_b^0 : $\psi_\gamma(0) = \alpha\psi_a^0 + \beta\psi_b^0$. From above it follows that either $\gamma = \mu$, in which case $\beta = \langle \psi_b^0 | \psi_\gamma(0) \rangle = 0$, and the good state is simply ψ_a^0, or $\gamma = \nu$ and the good state is ψ_b^0.

Once we identify the "good" states by solving Equation 8, we can use these "good" states as our unperturbed states and apply ordinary non-degenerate perturbation theory. In most cases, the operator A is suggested by symmetry, as symmetries are associated with operators commuting with the Hamiltonian – precisely what we need.

Moral of the story – If you are faced with degenerate states, look for a hermitian operator A that commutes with H^0 and H'. Pick you unperturbed states as ones that are simultaneously eigenfunctions of H^0 and A (with *distinct* eigenvalues). Then use ordinary *non*degenerate perturbation theory.

3 Higher-order Degeneracy

In the case of an n-fold degeneracy, we look for eigenvalues of the $n \times n$ matrix

$$W_{ij} = \left\langle \psi_i^0 | H' | \psi_j^0 \right\rangle . \tag{10}$$

For three fold degeneracy, (with degenerate states $\psi_a^0, \psi_b^0, \psi_c^0$), the first order corrections to the energies E^1 are the eigenvalues of W, determined by solving

$$\begin{pmatrix} W_{aa} & W_{ab} & W_{ac} \\ W_{ba} & W_{bb} & W_{bc} \\ W_{ca} & W_{cb} & W_{cc} \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = E^1 \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} \tag{11}$$

and the good states are the corresponding eigenvectors

$$\psi^0 = \alpha\psi_a^0 + \beta\psi_b^0 + \gamma\psi_c^0.$$

Again, if you can find an operator A that commutes with H^0 and H', and use the simultaneous eigenfunctions of A and H^0, then the W matrix will *automatically be diagonal*, and you won't have to calculate any off-diagonal entries of W or solve the characteristic equation.